NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMMERCE

DEPARTMENT OF ACCOUNTING

MANAGEMENT ACCOUNTING II

CAC 4105

QUESTION BANK

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PART 4.1 REVISION 2021

MANAC II

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Part 4.1 Tutorial Questions

Question 1 – Marginal and Absorption costing.

The following information is available for a firm producing and selling a single product:

Budgeted costs (at normal activity):

	$
Direct materials and labour	26 400 000
Variable production overhead	4 800 000
Fixed production overhead	14 400 000
Variable selling and administration overhead	2 400 000
Fixed selling and administration overhead	9 600 000

•The overhead absorption rates are based upon normal activity of 240 000 units per period.

•During the period just ended 260 000 units of product were produced, and 230 000 units were sold at $300 per unit.

•At the beginning of the period 40 000 units were in stock. These were valued at the budgeted costs shown above.

•Actual costs incurred were as per budget.

•Any under lower-recovery of overheads is set-off against cost of sales.

REQUIRED		**Marks**
(a)	Based on normal activity, calculate the unit product cost under both the absorption and marginal costing methods.	(3)
(b)	Calculate the fixed production overhead absorbed during the period, and the extent of any under/over absorption, if an absorption costing method is employed.	(2)
(c)	Calculate the operating profit for the period using absorption costing.	(9)
(d)	Calculate the operating profit for the period using marginal costing.	(9)
(e)	Reconcile the profit figures which you have calculated under (c) and (d) above.	(2)

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Question 2 – Activity Based Costing (32 marks)

Sweet Roast Limited is a producer of a variety of high grade coffee blends. The firm buys coffee beans mainly from the Ethiopian Highlands. The coffee beans are then roasted, blended and packaged in one kilogram bags for sale to specialist coffee houses around the world. The major raw material in the coffee blends is the beans. The company's automated roasting, blending and packaging processes result in substantial overheads. Since the company is highly automated it uses very little direct labour.

Coffee blends from the mainly Ethiopian Highlands are very popular and sell in large volumes while blends from Tanzania and Malawi sell in moderate volumes. The company has a mixed pricing strategy.

The company's budget for the coming year includes estimated production overheads of $2 200 000. The company assigns production overheads to products on the basis of direct labour hours. The expected direct labour cost is $600 000 and the budgeted directed labour hours are 50 000. Based on the sales budget and the material requirement budget, the company will, during the forthcoming year, purchase and use $5 000 000 of raw materials, mostly in the form of coffee beans.

The budgeted costs for direct materials and direct labour for one-kilogram bags for two of the company's coffee products are as follows:

	Ethiopian Roast	Malawi Roast
	$	$
Direct materials (per 1 kg bag)	4,50	2,90
Direct labour (per 1 kg bag) (0,02 Hours per bag)	0,24	0,24

The company's marketing manager is of the belief that the company's traditional costing system may be providing misleading cost information related to the products. She has learnt of alternative methods of assigning manufacturing overheads to products and has collected the following information for the **total production** in the company:

Cost pool	**Measure**	**Yearly Activity of all products**	**Budgeted Annual Cost**
			$
Purchasing	Purchase orders	2 000 Orders	560 000
Material Handling Costs	Number of set-ups	1 000 Setups	193 000
Quality control	Number of batches	500 batches	90 000
Roasting	Roasting hours	95 000 hours	1 045 000
Blending	Blending hours	32 000 hours	192 000
Packaging	Packaging hours	24 000 hours	120 000
Total manufacturing overhead			2 200 000

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Data regarding the expected production for **the two selected products** is given below:

	Ethiopian Roast	**Malawi Roast**
Forecast sales and production quantities)	80 000 kg	4 000 kg
Batch size	5 000kg	500 kg
Setups	2 per batch	2 per batch
Purchase order size	20 000 kg	500 kg
Roasting, blending and packaging time per 100 kg	2,3 hours	2,3 hours

The company holds no inventory of unroasted coffee beans at either the beginning or the end of the financial year.

REQUIRED		**Marks**
(a)	Using direct labour hours as the base for assigning manufacturing overhead cost to products, undertake the following: (i) Calculate the predetermined overhead recovery rate that will be used during the forthcoming year. (ii) Calculate the unit product cost of one kilogram each of the **Ethiopian and Malawi roast** products.	(1) (3)
(b)	Using activity based costing as a basis for assigning manufacturing overhead cost to products, undertake the following: (i) Calculate the total amount of manufacturing overhead cost that should be assigned to the total production of each of the roast blends for the year. (ii) Using the data calculated under b(i) above, compute the overhead cost per kilogram of each of the Ethiopian and Malawi roasts. (Work to 2 decimal places) (iii) Calculate the unit product cost of one kilogram each of the Ethiopian and Malawi roast products	(19) (3) (3)
(c)	In bullet point form, prepare short notes for the marketing manager on your findings under (a) and (b) above and clearly indicate the implications to the company of using direct labour as the base for assigning manufacturing overheads to products.	(5)

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QUESTION 3 – RELEVANT COSTING

Phumelela Engineering Ltd is a company based in Mbombela. It tenders for various size contracts in most parts of South Africa.

It has recently successfully tendered for a Local Authority contract in Mangaung in the Free State province. Work is due to begin in January next year. The price that the company submitted with which it won its bid was R432 000 000.

The company has also been requested to undertake another contract in Bela Bela in the Limpopo province. The price that has been offered for this work is R528 000 000. However, due to staffing problems, the two contracts cannot both be carried out at the same time. Phumelela has to choose which is the most financially rewarding to the company.

Phumelela is able to withdraw from the contract in Mangaung, thanks to a clause that was written into the contract before it was signed. However, if this is done the company would have to notify the Local Authority of its intention by October this year and pay a penalty clause of R42 000 000.

The following estimates relating to the costs and revenue of each contract have been compiled:

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	Mangaung $ 000	Bela Bela $ 000
Materials: Material A (in stock, original cost)	32 400	
Material B (in stock, original cost)		37 200
Committed orders at original cost for material A	45 600	
Further material A required (current cost)	90 000	
Further material C required (current cost)		106 800
direct labour wages from local agency	129 000	165 000
Plant on site: depreciation	14 400	19 200
Site foremen	51 000	51 000
Temporary accommodation for foremen	10 200	8 400
Interest cost at 8%	7 680	9 600
Total contract costs	**380 280**	**397 200**
Central overheads from headquarters	19 014	19 860
Total costs	**399 294**	**417 060**
Contract price	432 000	528 000
Anticipated profit	**32 706**	**110 940**

Additional information is as follows:

1. Material A does not have a resale value outside of the company, as it is not used commonly in the engineering industry. It could be used as a substitute for another material which currently costs 10% less than the original cost of material A.

2. The price for material B has increased to twice the amount at which it was first purchased. Material B can be resold at its current market price, in which case the costs of disposal are 12% of the selling price. If not sold, any material B could be used on any future contracts as it does not deteriorate.

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3. The plant that would be required is common to both contracts. If it were used on the Bela Bela. The plant that would be required is common to both contracts. If it were used on the Bela Bela contract, all of the plant would be required due to the nature of the work to be undertaken.
   However, if the Mangaung contract goes ahead, less plant will be required. The plant that is not used will be rented to a neighbouring business for a total rental of R9 000 000 during the period of the contract.

4. The interest charge of 8% relates to a notional charge for working capital that may be required during the period of the contract. The customer would pay the company progress payments according to the work completed.

5. Both contracts would run for a 12-month period.

6. Site foremen are paid an annual salary. This does not vary according to the amount of work undertaken. They will however, be expected to stay in temporary accommodation near the contract site. The accommodation costs have been estimated based on past experience.

7. The costs of the central headquarters consist of the salaries of the personnel who work in the administration and payroll departments, which amount to R162 000 000 in total. The headquarters supervise approximately 15 contracts at any one time.

8. It is not expected that labour will be in short supply.

REQUIRED		**Marks**
(a)	Making use of the principles of relevant costing, prepare a statement that compares the relevant net benefits of the two contracts in Rand terms.	(14)

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QUESDTION 4

TOPICS- (STANDARD COSTING, CVP-ANALYSIS AND RELEVANT COSTING)

Central Beverages Trading as Delta Beverages recently introduced two premium products for its high end market, Novida and Super Sip.

The management accountant has provided the following budgeted production information for the month of July 2021.

	Novida	**Super Sip**
Direct labour hours per litre	**0.20 hours**	**0.28 hours**
Budgeted Units to be produced	**5 500 litres**	**3400 litres**

In the month of July, 55 workers were employed in the production process on a government approved standard 40-hour week for the Beverage Industry. The standard direct labour rate for these employees was ZWL $10.00 per hour. Glucose is one of the direct materials used in the manufacture of both beverages. The Management Accountant has provided the following actual results for the month of July:

Quantities Manufactured

Novida	5 400 litres
Super Sip	4 100 litres
Direct labour paid	$ 88 700.00
Glucose purchased	7230 litres costing $ 13 095
Glucose Price Variance	$ 2 088.00 (Favourable)
Glucose Used (Novida)	5 142.50 litres
Glucose Usage Variance	$ 1 613.85 (Adverse)

The management accountant has also provided the following budgeted costs and revenues per litre for both beverages for the year ending 31 July 2022.

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	Novida	Super-Sip
	$	$
Selling Price	14.60	6.75
Variable Costs	8.05	3.60
Fixed Costs:		
Product Specific	2.45	1.75
General	2.30	2.30

The “per litre” revenue and cost data provided above is based on the following annual sales and production volumes for the year ending 31 July 2022.

Novida	**71 500 litres**
Super-Sip	**44 200 litres**

Ingutsheni Central Hospital has made a **“<u>once-off order</u>”** for the next financial year for 40 000 litres of a health sugar free beverage called Fanta-Grape for their patients. The per litre costs of manufacturing Fanta-grape are given as follows**:**

	ZWL $
Variable costs (excluding Caramel- refer below	**10.80**
Fixed Costs:	
Product Specific	**7.80**
General	**9.60**

Excluded in the Variable costs is direct material Caramel. Fanta Grape requires 0.50 litres of Caramel per Litre. The warehouse manager has confirmed that Caramel in inventory (8 000 litres) is not used in production of any current products and if not used soon will have to be disposed at a cost of $ 1.00 per litre. It was originally purchased at $8.00 per litre. The current prevailing market price of Caramel is 9,40 per litre.

For this special order Delta will have to hire additional labour to assist in administration and distribution at a cost of $31 000.00.

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	Required	Marks
(a)	Calculate the direct labour variances for the month of July	(6)
(b)	Calculate the standard purchase price for glucose for the month of July	(2)
(c)	Calculate the standard usage of Glucose per litre of Novida.	(3)
(d)	Based on the budgeted revenue mix calculate the break-even point revenue (ZWL) for Delta Beverages.	(10)
(e)	Find the number of litres of Super-Sip at the break-even point determined in (d) above.	(3)
(f)	For the beverage Super-Sip comment on the viability of its production and sales.	(6)
(g)	Calculate the minimum selling price per litre of Fanta-Grape that Delta Beverages will have to charge to break even on the once-off order of Ingutsheni Central Hospital. Briefly comment on the choice of relevant costs	(10)

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QUESTION 5

K&S LTD; SAVE LTD

K & S Ltd manufactures floor polish and liquid soaps for commercial clients. Both product lines are sold in 10 litre containers. The company is critically aware of their trading environment and uses a standard costing system, which is reviewed every six months. The last review was done on 1 March 2013 for the six months ending 28 August 2013.

Floor polish division

The following standard cost was set for one 10-litre container of polish:

	Standard Cost $	Unit Cost $
Direct materials:		
5 litre policloro	$3,60 / litre	18
3,5 litre oxotone	$2,20 / litre	7,7
4 litre lather	$0,60 / litre	2,4
Direct labour: 0,2 productive hours	$24,60 / hour	4,92
Variable overhead: 0,2 hours		14,4
Fixed overhead		8
Container		5,1
Total Standard cost per unit		60,52
Sales price per unit		90

Budgeted idle time of 4% was considered in detailing the above standard cost card. P & S allocates both variable and fixed overhead costs to production on the basis of productive labour hours. The budgeted fixed manufacturing overhead for the six months is $10 320 000. Manufacturing activities and incurring of production cost take place evenly throughout a year.

The actual costs incurred during March 2013 amounted to:

Purchase of direct materials:

Policloro	1 100 000 litres	@ $3,80 / litre
Oxotone	725 000 litres	@ $2,30 / litre
Lather	830000 litres	@$0,56 / litre
Labour	450 000hours	@ $24,50 / hour
Variable Overhead	$3 072 800	
Fixed Overhead	$1 689 200	
Containers:	212 000 containers @ $5,10 each	

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The actual production for March 2013 was 210 000 units. All containers purchased were issued. 1 000 000 litres of Policloro, 790 000 litres of Oxotone and 850 000 litres of Lather were issued to production.

REQUIRED		**Marks**
(a)	For the Floor polish division, calculate the • Material • Labour Variances for March 2013 in as much detail as possible.	(8)
Total		8

Question 6 – ABC COSTING AND TARGET COSTING

Feline Pixels (Pvt) Ltd provides the following information concerning its two product lines:

	Product A	**Product B**
Number of units produced and sold	550	110
Machine setup hours per production run	7	17,5

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Product units manufactured per production run	110	55
Direct labour hours per unit	2	2

Total budgeted machine setup-related cost – R42 000

(a)	Calculate how much machine setup-related cost is allocated to each of A and B's product lines if Feline Pixels (Pty) Ltd uses a traditional cost allocation system based on direct labour hours. Calculate the fixed overhead cost per unit for Product A and B;	(2)
(b)	Calculate how much machine setup-related cost is allocated to each of A and B's product lines if Feline Pixels (Pty) Ltd uses activity-based costing (ABC). Calculate the fixed overhead cost per unit for Product A and B;	(2)
(c)	Indicate if overhead allocation to A and B's product lines is evidence of product diversity or volume diversity.	(1)
(d)	Explain how ABC will affect the break-even sales value of the two product lines.	(3)
Total		8

QUESTION 6 – continued

Teddy Ltd manufactures a range of similar teddy bears. Teddy Ltd is considering the following proposals for changes to the design of the teddy bears:

1. Reduce some of the decorations on the teddy bear e.g. stitching
2. Instead of using glass eyes, use plastic eyes (two eyes per teddy bear)
3. Replace the synthetic material that is currently used for filling the teddy bears with scrap material left over from the manufacturing of the body.

The following information has been obtained regarding the proposed changes:

a) The current glass eyes cost R180 per hundred while the plastic eyes will cost R140 per hundred. The plastic eyes are however not as strong as the glass eyes and are more likely to be damaged during the manufacturing process. It is estimated damaged and broken plastic eyes will be 10% of the quantity issued from stores compared to 5% of issues of glass eyes at present.

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b) Currently, the synthetic filling material is bought for R720 per tonne. One tonne is sufficient for filling 2 000 teddy bears.

c) Before the scrap material can be used to fill the teddy bears it will have to be shredded. This will cost R0,50 per teddy bear. There will be sufficient scrap material available for filling all the bears. The scrap material is currently disposed of at a cost of R1 000 per month.

d) It is expected that the reduction of the decorations on the teddy bear will reduce the attractiveness of the teddy bear and will result in an estimated drop in sales of 10% from the current level. It will also reduce production costs by R6,00 per teddy bear. It is not expected that the change in eyes or filling material will negatively affect sales volume.

e) 300 000 teddy bears are currently manufactured and sold per year. Apportioned fixed costs amount to R4 050 000 per year, which will not be impacted by the changes considered. The net profit per teddy bear at the current sales level is R30.

(e)	Assuming that all three proposals are implemented, prepare an analysis that shows the estimated effect on annual profit and which enables management to check whether the proposals combined will achieve an annual target profit increase of R250 000. The proposals for plastic eyes and the change in filling should be evaluated after the decoration reduction proposal has been evaluated.	(12)
(f)	Calculate the percentage reduction in sales due to the decoration reduction at which the implementation of all three design change proposals combined would result in the same total profit from the teddy bears as that earned before the implementation of the design changes.	(5)
Total		17

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QUESTION 7 - STANDARD COSTING – (35 MARKS)

Month 1

Matojeni Ltd processes and sells brown rice. It buys unprocessed rice seeds and then, using a relatively simple process, removes the outer husk of the rice to produce the brown rice. This means that there is substantial loss of weight in the process. The market for the purchase of seeds and the sales of brown rice has been, and is expected to be, stable. Matojeni Ltd uses a variance analysis system to monitor its performance.

There has been some concern about the interpretation of the variances that have been calculated in month 1.

1. The purchasing manager is adamant, despite criticism from the production director, that he has purchased wisely and saved the company thousands of dollars in purchase costs by buying the required quantity of cheaper seeds from a new supplier.
2. The production director is upset at being criticised for increasing the wage rates for month 1; he feels the decision was the right one, considering all the implications of the increase. Morale was poor and he felt he had to do something about it.
3. The maintenance manager feels that saving $8 000 on fixed overhead has helped the profitability of the business. He argues that the machines' annual maintenance can wait for another month without a problem as the machines have been running well.

The variances for month 1 are as follows:

		$
Material price	Favourable	48 000
Material usage	Adverse	52 000
Labour rate	Adverse	15 000
Labour effeiciency	Favourable	18 000
Labour idle time	Adverse	12 000
Variable Overhead expenditure	adverse	18 000
Variable overhead efficiency	Favourable	30 000
Fixed overhead expenditure	Favourable	8 000
Sales price	Adverse	85 000
Sales volume	adverse	21 000

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Matojeni Ltd uses labour hours to absorb the variable overhead:

Month 2

In month 2 the following standard data applies:

Standard costs for 1 tonne of brown rice:

- 1,4 tonnes of rice seeds are needed at a cost of $60 per tonne
- It takes 2 labour hours of work to produce 1 tonne of brown rice and labour is normally paid $20 per hour.
- 2 hours of variable overhead at a cost of $30 per hour
- The standard selling price is $240 per tonne
- The standard contribution is $56 per tonne

Budget information for month 2 is:

- **Fixed costs were budgeted at $210 000 for the month**
- **Budgeted production and sales were 8 400 tonnes**

The actual results for month 2 were as follows:

- Actual production and sales were 8 000 tonnes
- 12 000 tonnes of rice seeds were bought and used costing $660 000
- 15 800 labour hours were paid for costing $303 360
- 15 000 labour hours were worked
- Variable production overhead cost $480 000
- Fixed costs were $200 000
- Sales revenue achieved was $1 800 00

REQUIRED		**Marks**
(a)	Using the variances and other relevant information above comment on the performance of the following managers in month 1: i. The purchasing manager ii. The production director **iii.** The maintenance manager	**(10)**
	Note: You are not required to comment on the performance of the business or its managers in month 2.	
(b)	Calculate the variances for month 2 in as much detail as the information above allows.	**(10)**

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(c)	Reconcile the budget profit to the actual profit using marginal costing principles.	**(15)**

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